

Results Gallery

Contents

- Hazard
- Probability and Severity
- Vulnerability
- Risk
- Notes

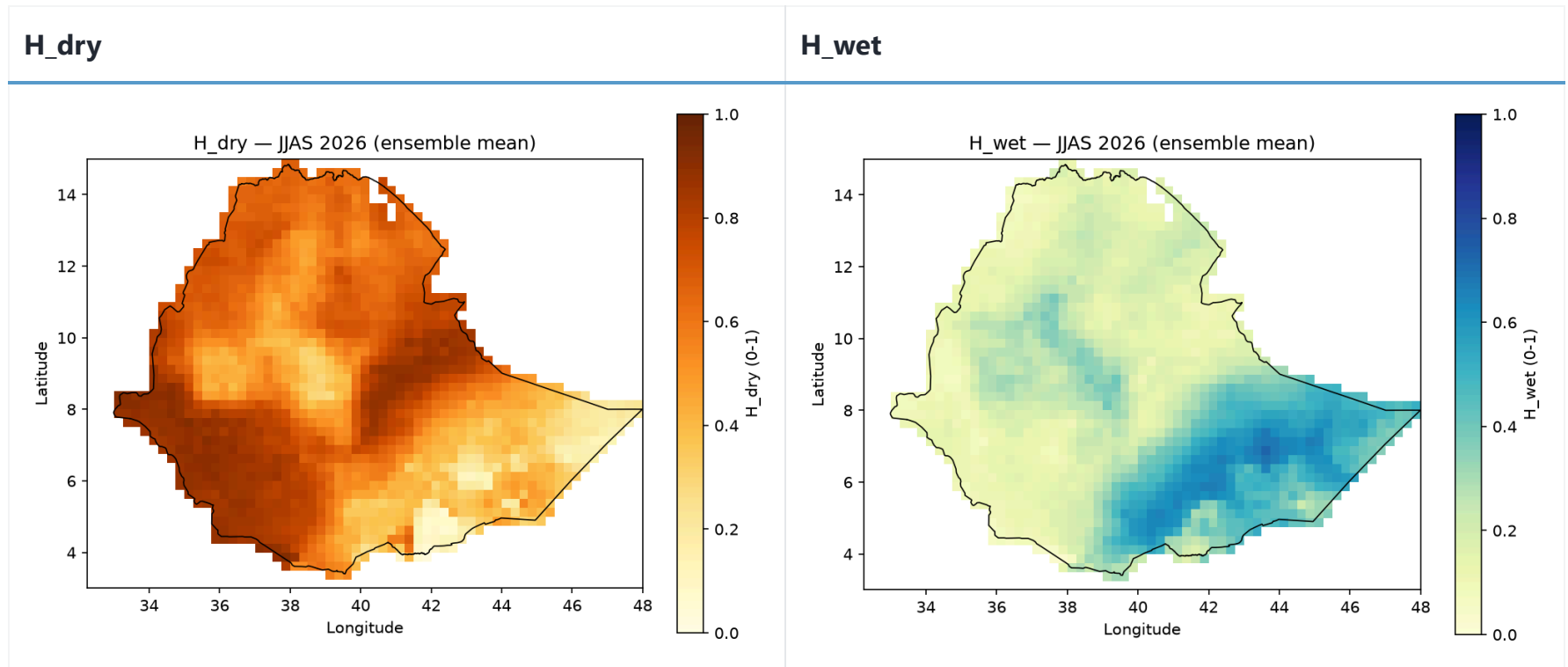
Representative outputs from the Ethiopia seasonal run, with the formula and data behind each stage shown next to its result. Full derivations and weights are in [methodology.md](#); component-level detail (data sources, sub-indicator breakdowns) is in [hazards.md](#), [exposure.md](#), and [vulnerability.md](#).

Hazard

$$H_{dry} = 0.35S_{SPI,dry} + 0.20S_{rain,dry} + 0.30S_{CDD,dry} + 0.15S_{CWD,dry} \quad H_{wet} = 0.20S_{SPI,wet} + 0.20S_{rain,wet} + 0.20S_{CWD,wet}$$

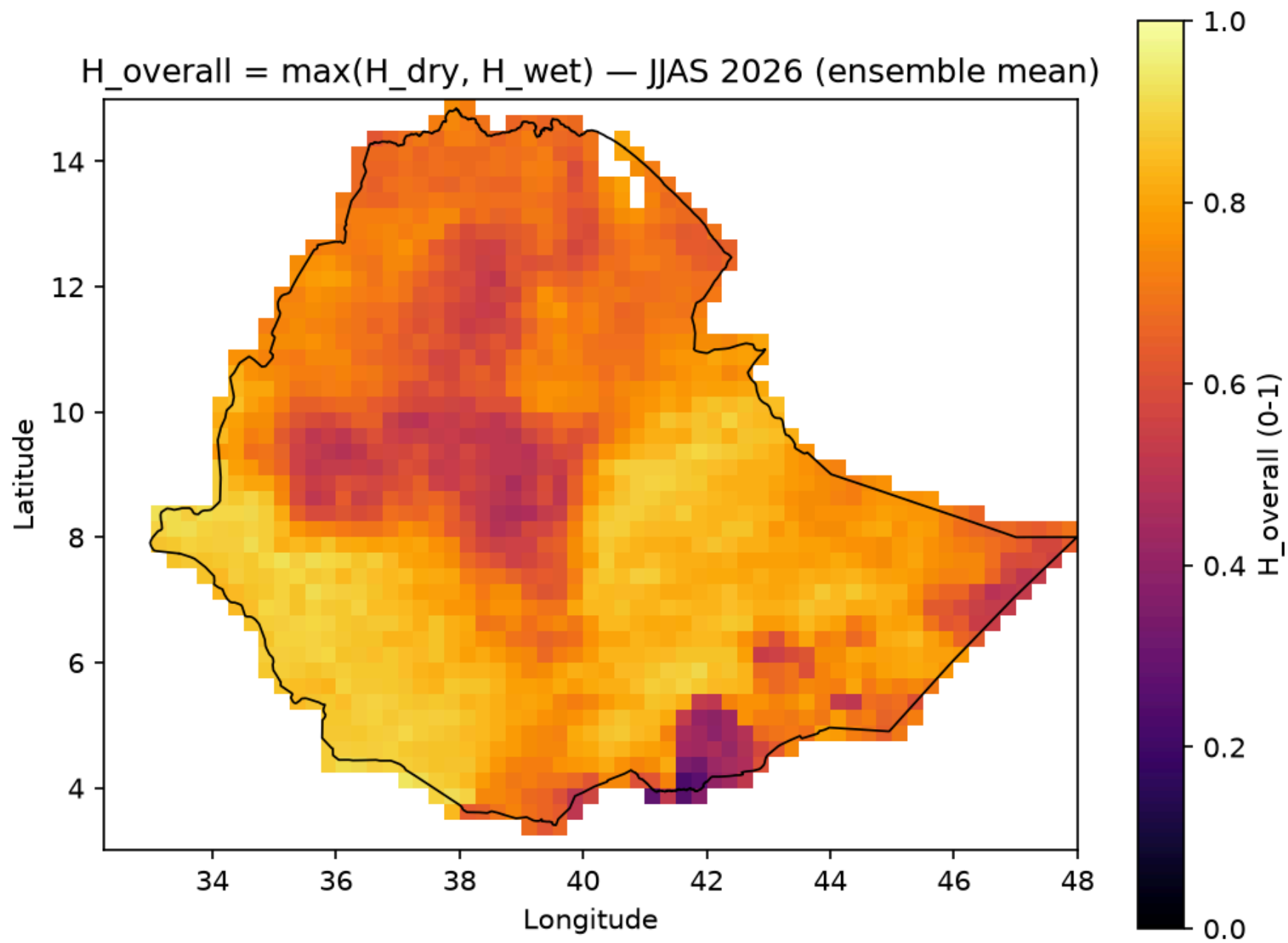
Data: SPI, rainfall percentile, CDD, CWD, Rx1day, Rx5day per ensemble member, from the sibling indicator-calculation project (see [data_provenance.md](#)).

Drought and excess wetness hazard (JJAS)



Combined hazard (max operator)

$$H_{overall} = \max(H_{dry}, H_{wet})$$



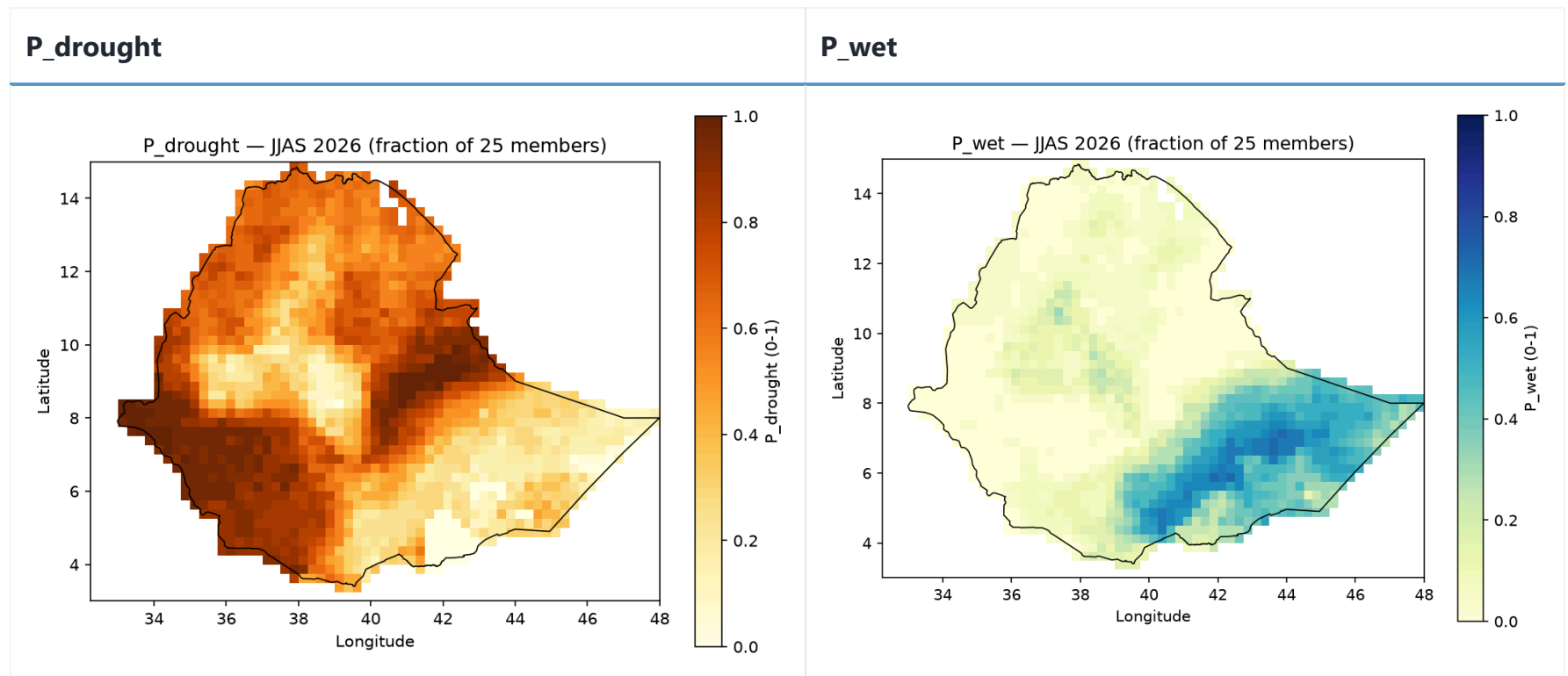
Probability and Severity

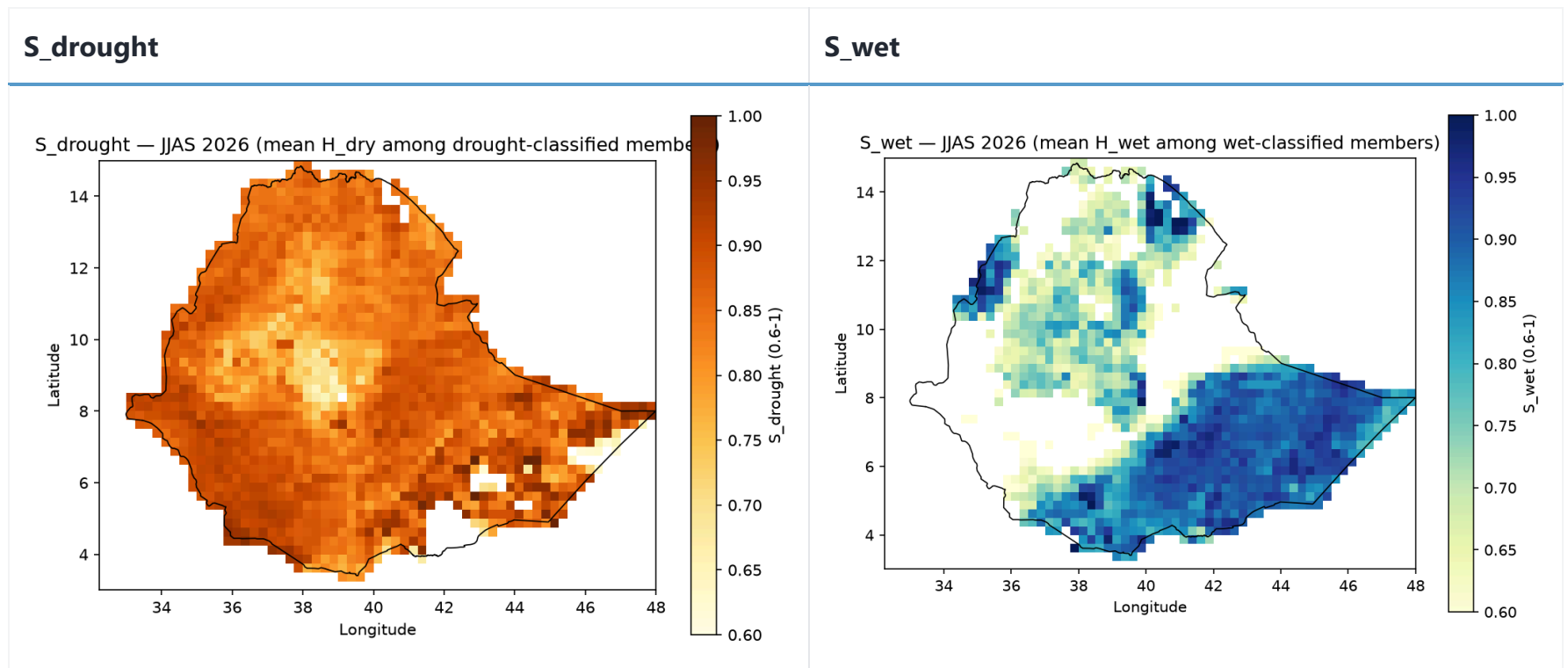
$$P_{drought} = \frac{\text{count}(H_{dry} \geq 0.60)}{\text{total ensemble members}}$$

$$S_{drought} = \text{mean}(H_{dry} \text{ among members classified as drought events})$$

(P_{wet} / S_{wet} are the mirror-image definitions using H_{wet} .) With this project's 25-member ensemble, probability moves in steps of $1/25 = 4\%$.

Data: same per-member H_{dry} / H_{wet} fields as above, collapsed across the ensemble dimension.





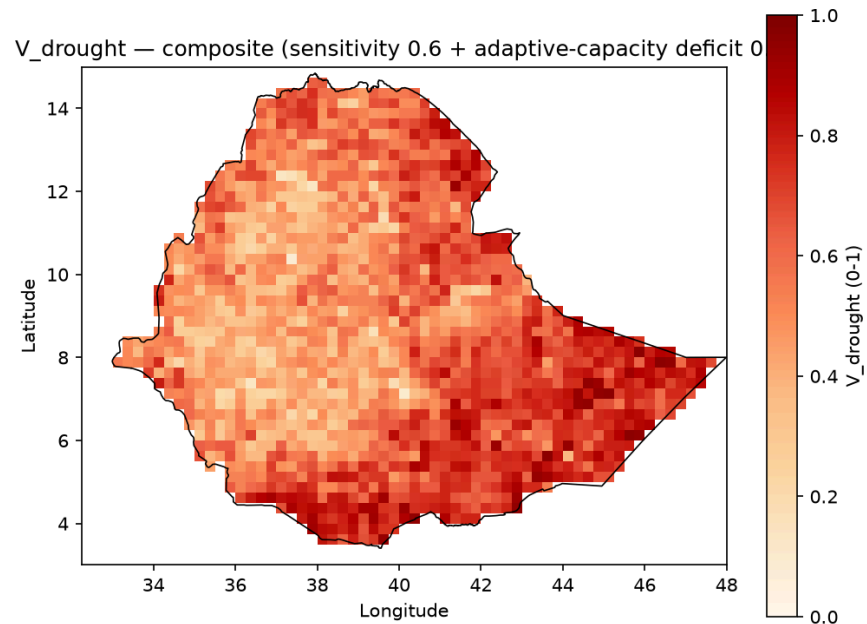
Vulnerability

$$V_{drought} = 0.60(0.5S_{poverty} + 0.5S_{aridity}) + 0.40(0.5S_{irrigation_deficit} + 0.5S_{no_electricity_access})$$

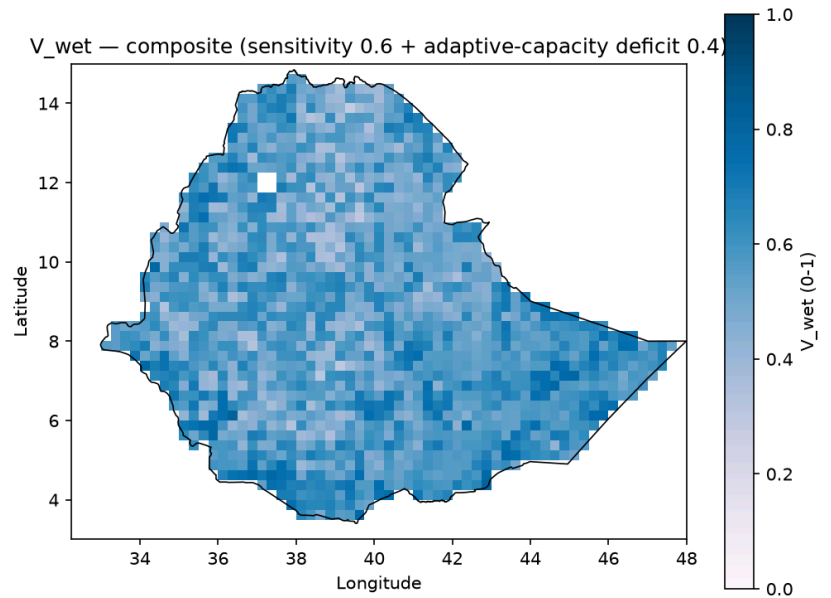
$$V_{wet} = 0.60(0.20S_{poverty} + 0.20S_{elevation} + 0.20S_{slope} + 0.20S_{soil_clay} + 0.20S_{river_distance}) + 0.40(0.5S_{healthsites_deficit} + 0.5S_{no_electricity_access})$$

Data: relative wealth index, aridity index, irrigated-area percent, population without electricity access, elevation, slope, topsoil clay content, distance to rivers/water bodies, health facility count – see [vulnerability.md](#) for the exact source per indicator.

V_drought



V_wet



Risk

$$R_{drought} = 100 \cdot P_{drought} \cdot S_{drought} \cdot E \cdot V_{drought} \quad R_{wet} = 100 \cdot P_{wet} \cdot S_{wet} \cdot E \cdot V_{wet}$$

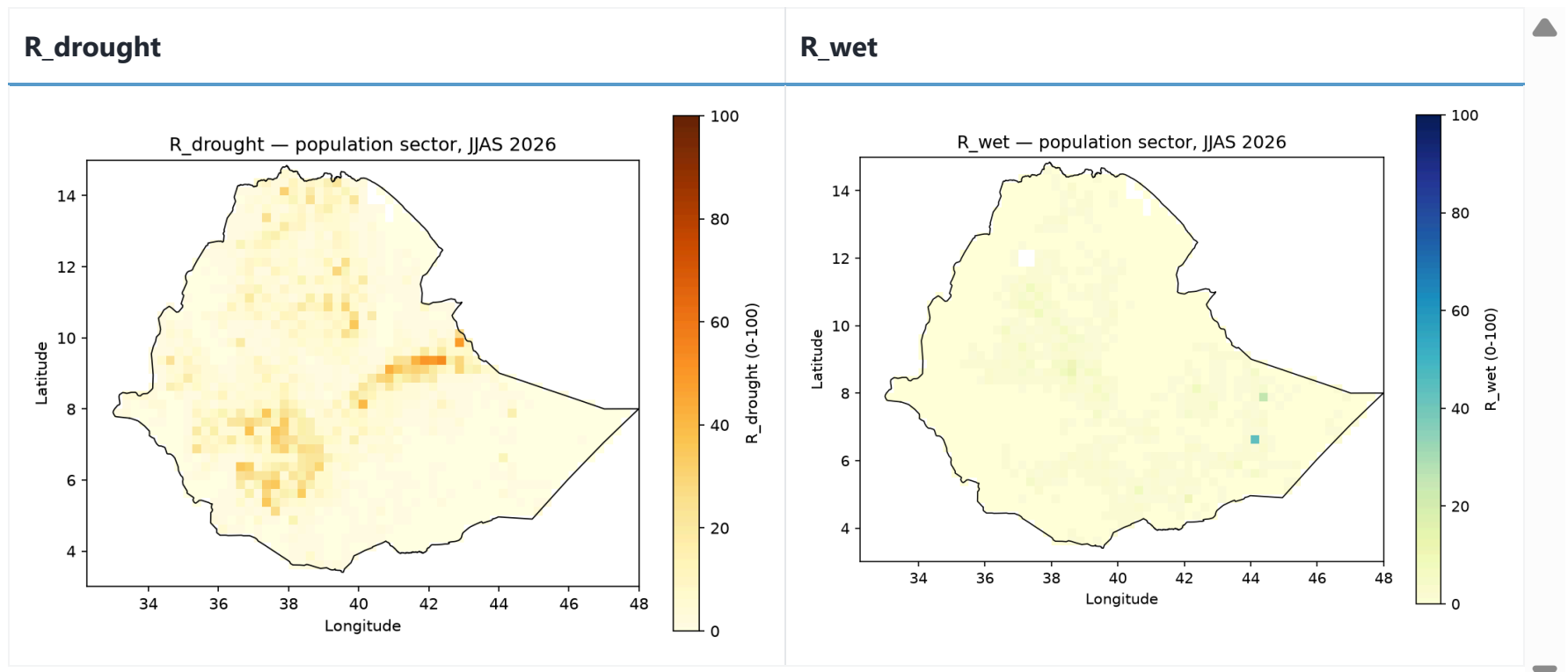
$$R_{dominant} = \max(R_{drought}, R_{wet})$$

R is a relative 0-100 score, **not** a probability percentage, computed per exposure sector (the figures below use population).

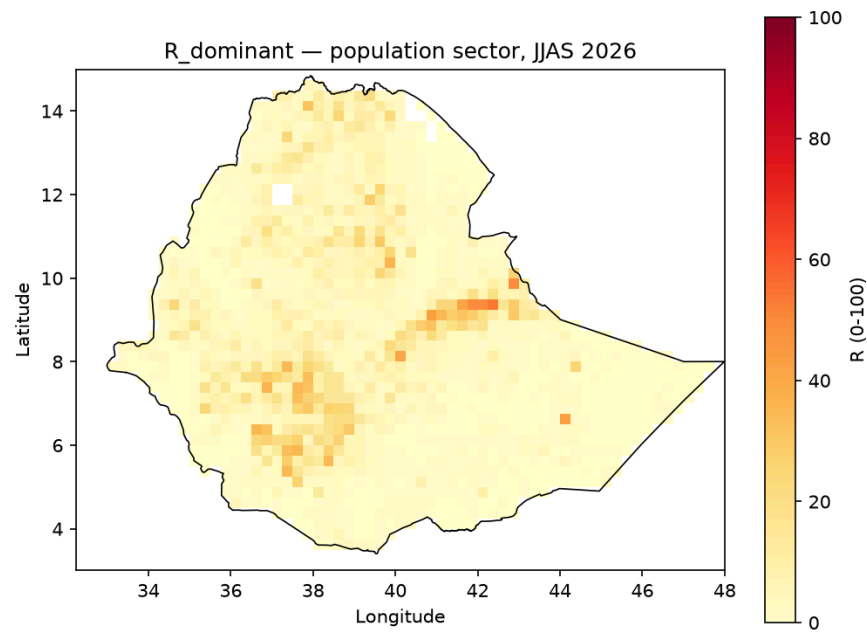
Risk class boundaries:

Range	Class
0-19.9	Very low
20-39.9	Low
40-59.9	Moderate
60-79.9	High
80-100	Very high

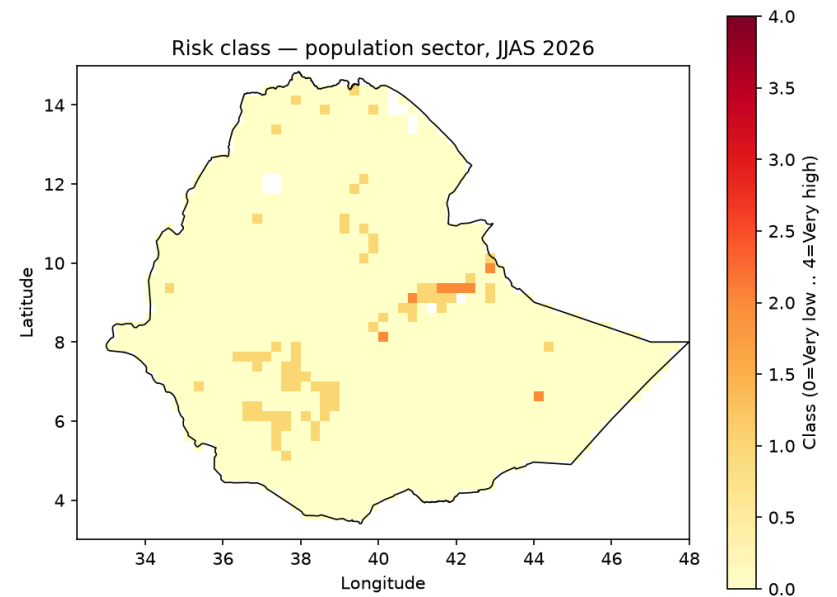
Data: P/S from the Hazard/Probability stage above, E from [exposure.md](#) (population sector, normalized), V from the Vulnerability stage above.



Dominant risk



Risk class



Notes

- These figures are rendered products intended for interpretation and communication.
- Full machine-readable outputs are in [outputs/geotiff/](#) and [outputs/netcdf/](#).
- Hazard composition and standardization details are documented in [methodology.md](#).